Department of Computer Science & Engineering
The University of Texas at Arlington

Detailed Design Specification
CSE 4317: Senior Design II
Summer 2025

EGG UTA
(Encapsulated Gardening Grounds)

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Revision History

Revision	Date	Author(s)	Description
0.1	06.17.2025	DD, SH, KM, CS, KT	Document creation
1.0	8.1.2025	DD	Document finalization

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1 Introduction

The Encapsulated Gardening Grounds, otherwise known as the EGG, is a project and product designed to assist individuals and small-scale growers maintaining crops more efficiently through a combination of an embedded system and an external app. This system is predicated on requisite hardware and software to assist these individuals, including various sensors and an associated web application to offer maintenance advice and manual control over various hardware automation.

More detailed information about the EGG's requirements and architecture can be found in the System Requirements Specification and the Architectural Design Specification, respectively.

2 System Overview

The overall structure of the EGG system includes four distinct layers: Sensors, Web Application and Software, Physical Materials and Build, and Hardware and Automations. The purpose of these layers is to address both the software and hardware features of the EGG whilst also accounting for data gathering and physical make-up of the EGG. The EGG needs to be able to read and store data, process user input, and physically house and attend to the plants using various embedded tools.

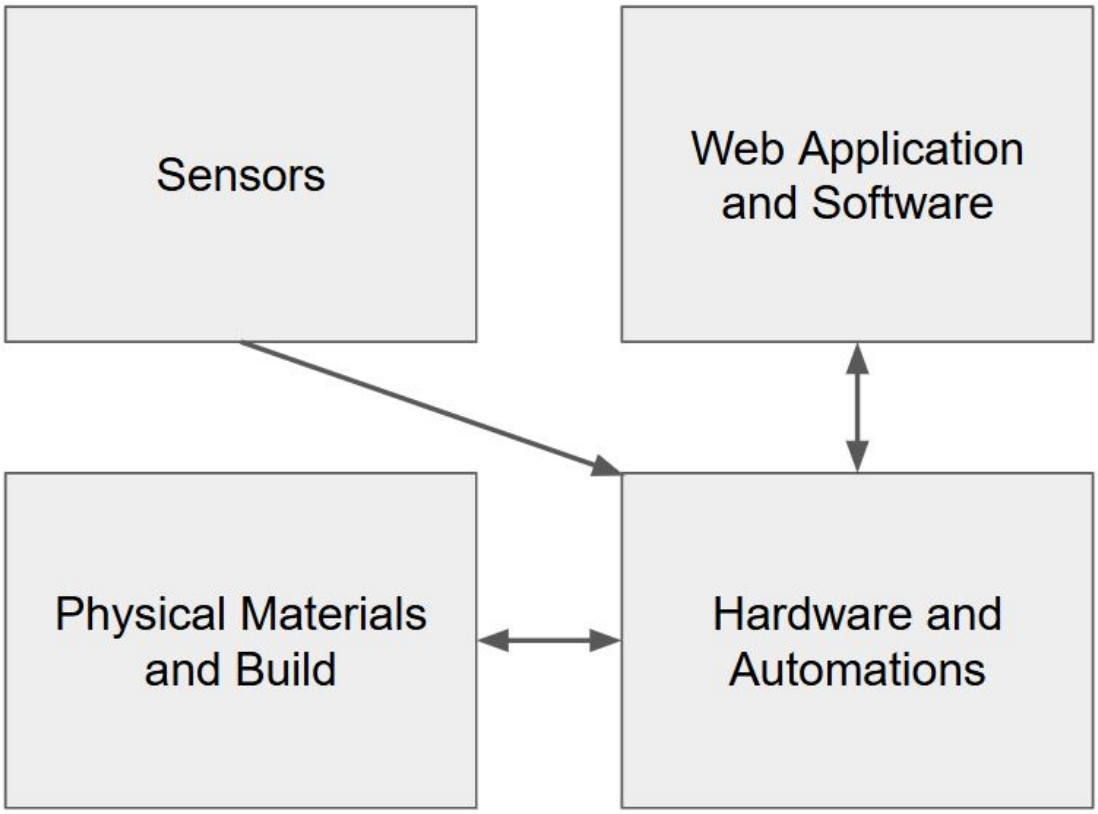

Figure 1: System Architecture

3 Sensor Layer Subsystems

The sensor layer consists of three types of sensors: moisture, nutrition, and humidity/temperature. These sensors collect data about the EGG environment and send it to the microcontroller.

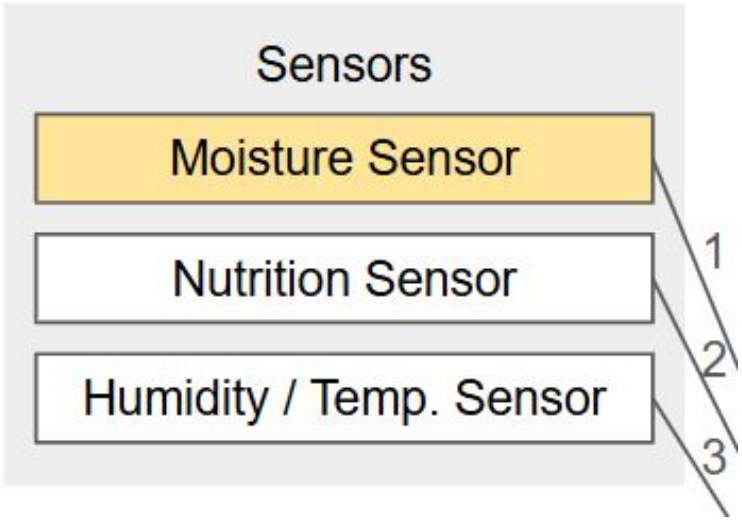

Figure 2: Moisture Sensor subsystem diagram

3.1 Subsystem: Moisture

The subsystem measures the moisture of the enclosure's soil and relays it to the microcontroller. The moisture sensor, a Hygrometer, in addition to a LM393 chip, sends an analog voltage a designated analog pin on the Arduino. The value of this voltage is then added to the logged values in the Arduino's hosted JSON log file.

3.1.1 Moisture: Hardware

As mentioned above, the moisture sensor is composed of Hygrometer and LM393 chip.

3.1.2 Moisture: Data Structures

The moisture sensor sends analog values to the microcontroller where they are converted to digital, and added to the Arduino's hosted JSON log file. This file is hosted locally via an SD Card Adapter connected to the Arduino, where it is retrieved and further worked upon by the local application.

3.2 Subsystem: Nutrition

A nutrition sensor will collect data concerning the health of the soil (the quantities of nitrogen, phosphorus, and potassium) in the EGG.

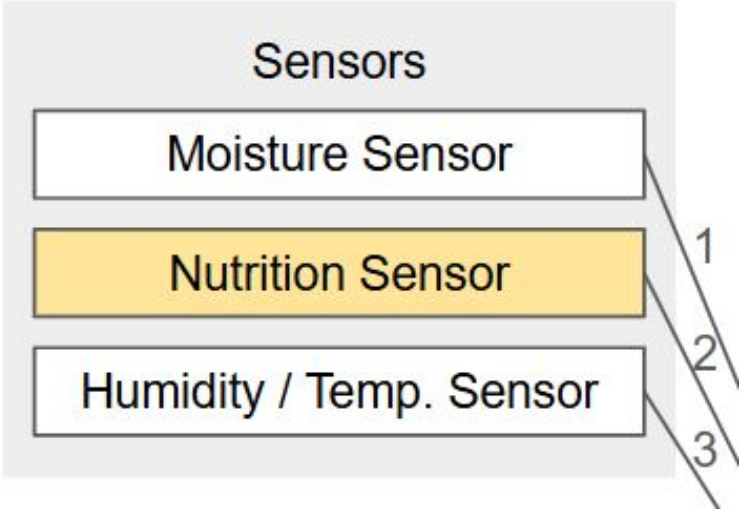

Figure 3: Nutrition Sensor subsystem diagram

3.2.1 Nutrition: Hardware

The sensor used to measure the nutrition levels in the soil of the Encapsulated Gardening Grounds is a JXCT Soil NPK Sensor. Specifically, our system uses the SEN0605 DFRobot NPK Sensor. The NPK sensor uses the RS485 communication protocol to output resulting data concerning the Nitrogen, Phosphorous, and Potassium content of the soil. This RS485 communication is translated into a usable format via the Arduino's serial communication (UART). The intention is to output these values to advise the user for proper care of their crop(s) in regards to the soil's overall nutrition.

3.2.2 Nutrition: Data Structures

As with the other sensors' parsed data, the Nitrogen, Phosphorous, and Potassium levels measured by the NPK Sensor are added to the Arduino's hosted JSON file as another column in the vast log file.

3.3 Subsystem: Humidity & Temperature

A combination humidity and temperature sensor is used to collect data about the relative humidity and temperature of the EGG's encapsulated space.

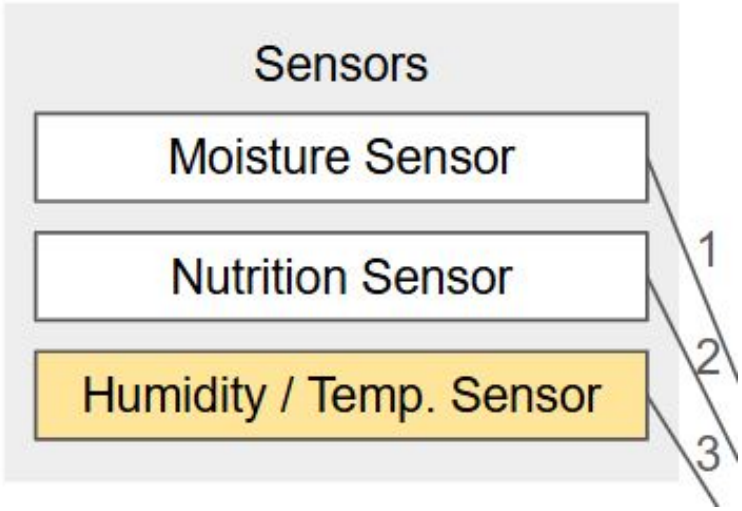

Figure 4: Humidity & Temperature subsystem diagram

3.3.1 Humidity & Temperature: Hardware

The combination humidity and temperature sensor used to get the respective values is the DHT22 sensor.

3.3.2 Humidity & Temperature: Software Dependencies

Within the Arduino's persistent program, a dedicated DHT22 Library is used to facilitate the humidity and temperature sensor's communication to the Arduino. The library is a part of the Adafruit Unified Sensor Library.

3.3.3 Humidity & Temperature: Data Structures

As with the other sensors' parsed data, the temperature and humidity levels measured by the DHT22 are added to the Arduino's hosted JSON file as another column in the vast log file.

4 Application and Software Layer Subsystems

The Application and Software Layer serves as the interface between the user and the underlying physical hardware of the automated plant grower system. This layer is responsible for managing user interactions, storing, and processing sensor data, as well as providing feedback through a user-friendly interface.

4.1 Application and Software Layer Operating System

Due to the cron scheduling implementation, the EGG application's needs to run on a Unix system, such as a Raspberry Pi.

4.2 Subsystem: Data Storage

The data storage subsystem stores sensor data fetched from the .json on a local database file. A python file fetches the .json sensor data from the microcontroller, and adds any new sensor data to the .db file in the same directory.

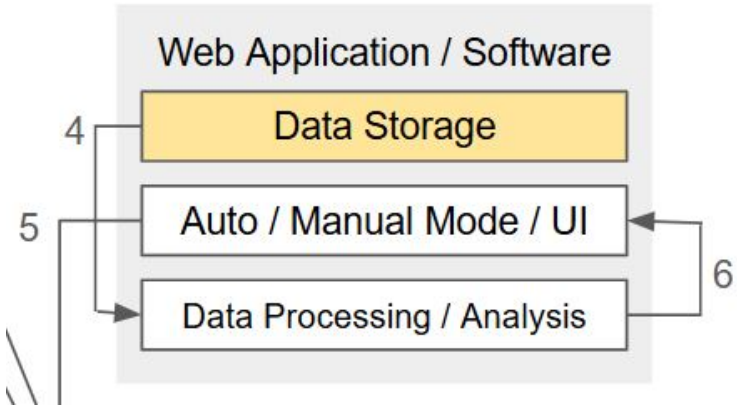

Figure 5: Data Storage subsystem diagram

4.2.1 Data Storage: Software Dependencies

The subsystem uses SQLite, a library that implements a small SQL database. It can be imported as sqlite3 from Python itself. It also borrows .json fetching code and the IP address from another backend file, interface.py.

4.2.2 Data Storage: Programming Languages

The code itself is Python3, while SQLite is written in C.

4.2.3 Data Storage: Data Structures

The sensor data is read into the program as a .json.

4.2.4 Data Storage: Data Processing

The program reads in the .json data, parses the fields, and uses a parameterized statement to execute the SQL. It will typically run regularly as a cron task.

4.3 Subsystem: Auto/Manual Mode/UI

The Auto/Manual Modal Control and UI subsystem is the primary means by which users interact with the automated plant grower system. It allows users to switch between automatic and manual control

modes, configure system preferences, and view real-time system status and environmental data. This subsystem acts as a bridge between the user and the rest of the web application and software layer.

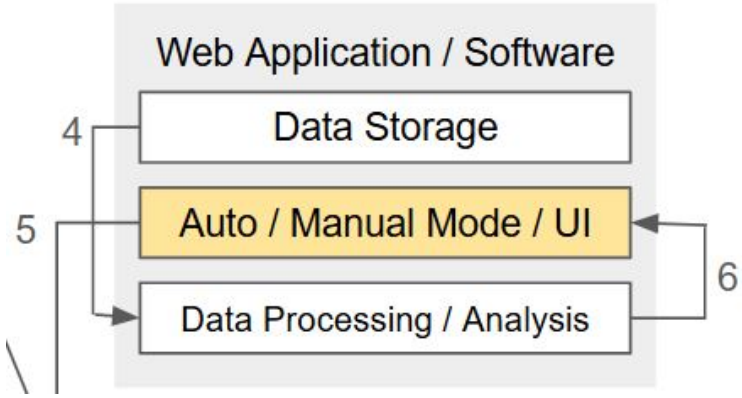

Figure 6: Auto/Manual Mode/UI subsystem diagram

4.3.1 Auto/Manual Mode/UI: Hardware

The subsystem runs on the host computer, which is a Raspberry Pi 4 that is running Raspberry Pi OS.

4.3.2 Auto/Manual Mode/UI: Operating System

The subsystem must run on Linux. This is because the application requires Linux's CronTab job scheduler to automate tasks.

4.3.3 Auto/Manual Mode/UI: Software Dependencies

Python Libraries

- Requests
- JSON
- SQLite3
- CronTab
- Flask

UI Tooling

- Vite + Electron
- React Library
- React Bootstrap
- Recharts

4.3.4 Auto/Manual Mode/UI: Programming Languages

- TypeScript (Frontend)
- Python (Backend)
- Bash (CronTab Scheduling)
- HTML/CSS (for UI)
- JSON (data format)

4.3.5 Auto/Manual Mode/UI: Data Structures

Sensor data is tracked and stored in a SQLite database and information for plant care will be sourced from a third-party API or database.

4.3.6 Auto/Manual Mode/UI: Data Processing

User-end requests, which are automated and manual plant care tasks, are sent via http request to the Arduino over local internet. Flask is used to link the React frontend with a Python backend. Automated tasks in particular will be sent to the Pi's CronTab scheduler, where the http request will be sent regularly at scheduled times.

4.4 Subsystem: Data Processing/Analysis

This subsystem handles receiving, storing, interpreting, and visualizing data from the EGG's microcontroller. It is primarily implemented as a set of web-based services and front-end components that fetch sensor data over HTTP, store it in a database, and display trends over time to the user via graphs and dashboards. It provides insight into environmental conditions and helps automate decisions (e.g., when to water or ventilate).

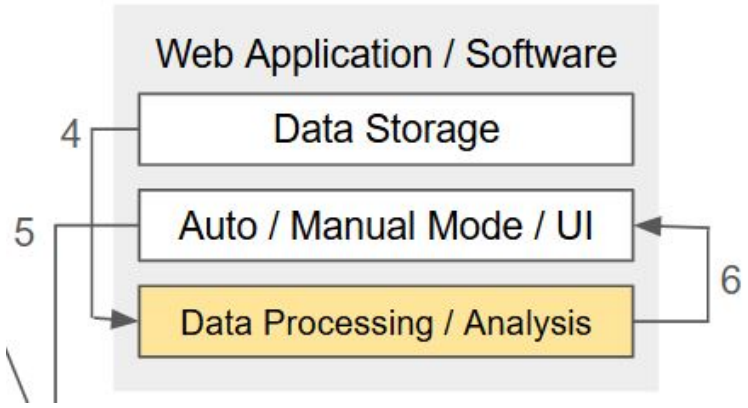

Figure 7: Data Processing/Analysis subsystem diagram

4.4.1 Data Processing/Analysis: Hardware

These items are a reiteration of the previously described sensors, and how they apply to the data handling, processing, and analysis of the EGG and its associated application.

- Soil Moisture Sensor: The values parsed from the moisture sensor are documented in an ongoing, hosted JSON file. Analysis on these values is performed by the application, which is retrieved by the Arduino's hosted HTML website.

- DHT22 Temperature and Humidity Sensor: The values parsed from the temperature and humidity sensor are documented in an ongoing, hosted JSON file. Analysis on these values is performed by the application, which is retrieved by the Arduino's hosted HTML website.
- Nitrogen, Phosphorous, and Potassium Sensor: The values parsed from the NPK sensor are documented in an ongoing, hosted JSON file. Analysis on these values is performed by the application, which is retrieved by the Arduino's hosted HTML website.

4.4.2 Data Processing/Analysis: Operating System

- Data from the hosted JSON is hosted on an HTML website, which is accessible from the majority of available operating systems. Our implementation of the data processing for this project utilizes a Raspberry Pi running a Unix-based Operating System.

4.4.3 Data Processing/Analysis: Software Dependencies

- **Web Server:** Flask (Python-based lightweight web framework)
- **Scheduler:** Cron (for scheduling periodic Python data fetch/processing jobs)
- **Data Analysis Scripts:** Custom Python scripts to parse and process sensor data
- **Desktop Application:** Electron.js (for UI and dashboard rendering)
- **Data Visualization:** Recharts (used in the Electron frontend to plot sensor trends)
- **Inter-process Communication:** REST API endpoints exposed by Flask, consumed by the Electron app

4.4.4 Data Processing/Analysis: Programming Languages

- TypeScript (Frontend)
- Python (Backend)
- Bash (CronTab Scheduling)
- HTML/CSS (for UI)
- JSON (data format)

4.4.5 Data Processing/Analysis: Data Structures

Sensor data is structured in JSON format. Each packet represents a snapshot of environmental readings collected by the Arduino and exposed via an HTTP endpoint served by the microcontroller.
Figure X: JSON data packet from microcontroller

```
{
  "temperature": 23.7,
  "humidity": 58,
  "soil_moisture": 295,
  "timestamp": "2025-05-22T15:54:39"
}
```

These packets are periodically fetched by a Python script running on a host machine. The script is triggered on a schedule using cron, processes the data if needed, and makes it available via a local Flask server. The Electron application queries this local Flask API (e.g., `http://localhost:5000/data`) to retrieve the latest sensor values, which are then visualized using Recharts.

4.4.6 Data Processing/Analysis: Data Processing

As briefly mentioned, the flow of data from the Arduino is first facilitated through an intermediary Raspberry Pi. The local application then interacts with this Pi for multiple features and requirements of the Encapsulated Gardening Grounds.

5 Physical Materials and Build Layer Subsystems

This section of the DDS details the subsystems that support the structural construction or mechanical operation of the device. Structural elements support the device's longevity and contain the encapsulated environment. Mechanical elements support plant maintenance systems, such as plumbing and water supply systems.

5.1 Subsystem: Water Reservoir/Drain Pan

The water reservoir provides water to the EGG's irrigation system, where a pump draws water into the plumbing network. After irrigation, any unused water is captured as runoff by the drain pan and returned to the reservoir. The drain pan employs a filtering process that will stop soil or other debris from returning to the reservoir. This allows the device to conserve water efficiently while protecting the plumbing system and keeping maintenance simple. The user is also able to refill the reservoir manually as needed. The materials used for this subsystem are nonreactive and durable for water and outdoor use.

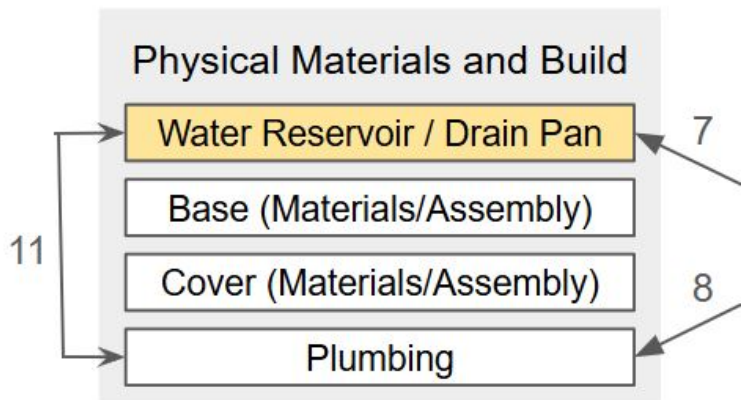

Figure 8: Water Reservoir/Drain Pan subsystem diagram

5.1.1 Water Reservoir/Drain Pan: Details

The water reservoir is made up of a combination of interior plywood and exterior paneling. The walls of wood make up the shape of this bottom assembly to act as the drain pan for the top assembly. In addition to a couple speed holes made where necessary, the drain pan is also lined with waterproof lining to allow for water to be held safely without damaging the wood.

5.2 Subsystem: Base (Materials/Assembly)

This subsystem contains the device components and gardening elements. Together with the cover subsystem, it encapsulates the gardening environment, separating the internal climate from external conditions. The materials used for this subsystem should be nonreactive and durable, suitable for exposure to water and outdoor environments.

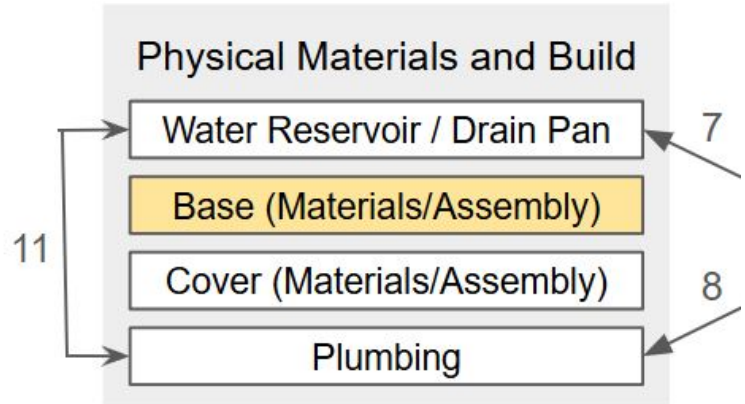

Figure 9: Base subsystem diagram

5.2.1 Base (Materials/Assembly): Details

The base is made up of a combination of interior plywood and exterior paneling. The walls of wood make up the shape of this bottom assembly to act as the container for the EGG's crop's soil. In addition to a couple speed holes made where necessary, the base is also lined with waterproof lining to allow for water and soil to be held safely without damaging the wood.

5.3 Subsystem: Cover (Materials/Assembly)

The cover will allow the user to see into the EGG while closed and act as a lid, allowing the user to open and access the plants and sensors.

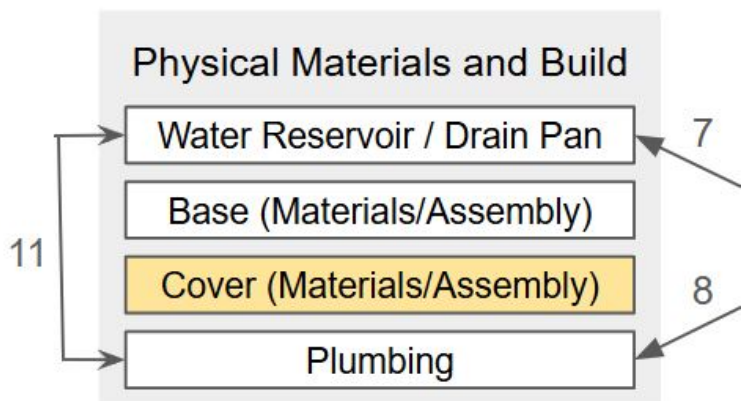

Figure 10: Cover subsystem diagram

5.3.1 Cover (Materials/Assembly): Details

The Cover is made up of 3D printed components to facilitate a smooth opening and closing of the lid. The cover provides an encapsulated area for the EGG's crops, and enables humidity control more closely. This enclosure is properly sealed with plexiglass walls, making the enclosure akin to a greenhouse.

5.4 Subsystem: Plumbing

The Plumbing subsystem enables the delivery of water from the reservoir to the plants and ensures safe transport of excess water to the drain pan. Plumbing consists of tubing, pumps, and connectors designed

to handle low-pressure water flow. The materials used for this subsystem should be nonreactive and durable for water use and the external environment.

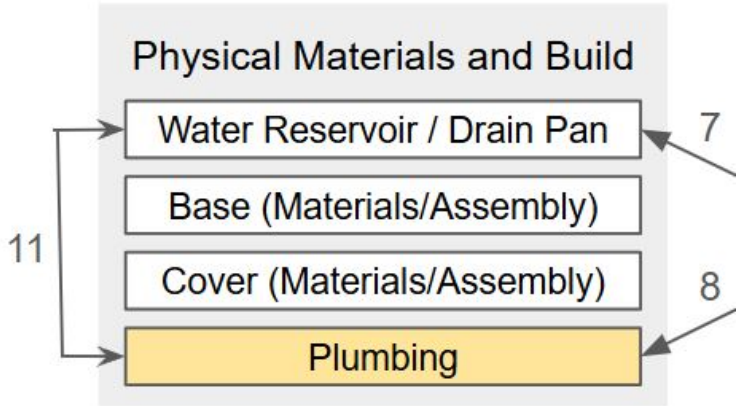

Figure 11: Plumbing subsystem diagram

5.4.1 Plumbing: Details

The plumbing system is simply comprised of clear tubing and a pump controlled by the Arduino. The tubing utilizes the aforementioned speed holes to connect the drain pan and facilitate water to the plant.

6 Hardware and Automation Layer Subsystems

This system encapsulates the majority of the non-structural hardware used the EGG. This includes all of the computing items and microcontrollers used to implement the Encapsulated Gardening Grounds.

6.1 Subsystem: Microcontroller

The Microcontroller serves as the primary data collector and sender for the EGG. The controller is directly connected to the various sensors in order to drive and sample them as needed. Additionally, the controller is to drive the EGG's automated systems, including the pump and watering system, and the fans for dehumidifying and temperature control. The controller will directly interact with the data storage system via an SD Card Adapter to keep data to be worked on and output by the associated desktop application. The selected controller for the Encapsulated Gardening Grounds is the Arduino UNO R4 WiFi.

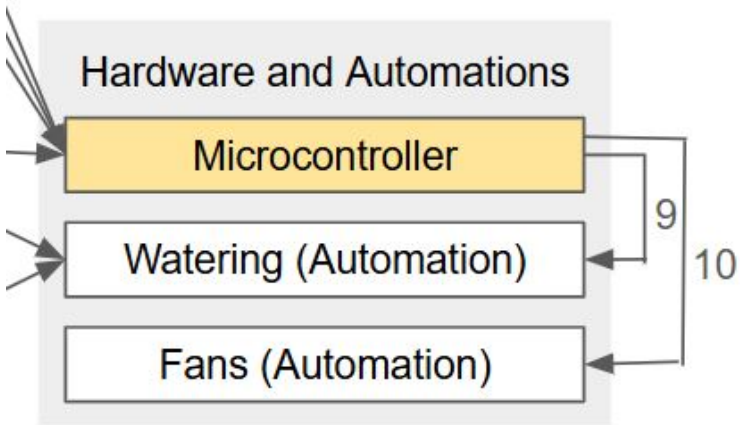

Figure 12: Microcontroller subsystem diagram

6.1.1 Microcontroller: Hardware

- **Microcontroller Board:** Arduino UNO R4 WiFi
- **Connected Sensors and other Hardware:**
  - Soil moisture sensor
  - DHT22 Temperature and Humidity sensor
  - NPK
  - Water Pump
  - Fans

6.1.2 Microcontroller: Software Dependencies

- Standard Arduino SD Card library
- Standard Arduino RTC library
- Standard Arduino WiFi library
- HT Sensor Library: https://github.com/adafruit/DHT-sensor-library
- Adafruit Unified Sensor Lib: https://github.com/adafruit/Adafruit-Sensor

6.1.3 Microcontroller: Programming Languages

- C/C++ (Arduino dialect)

6.1.4 Microcontroller: Data Structures

While not specifically integral to the microcontroller itself, our system utilizes an SD Card Adapter to interact with an SD Card. The SD Card hold the temporary log files to be used by the associated local application, and documents the timestamped values of the parsed sensors. This log file, and the data held within it, is updated every 5 minutes.

6.1.5 Microcontroller: Data Processing

- Sensor averaging
- Threshold detection for watering and ventilation
- Event triggers based on sensor values

6.2 Subsystem: Watering (Automation)

This subsystem is a hardware-software hybrid that controls the irrigation mechanism. It receives instructions from the microcontroller to activate the water pump when soil moisture drops below a defined threshold.

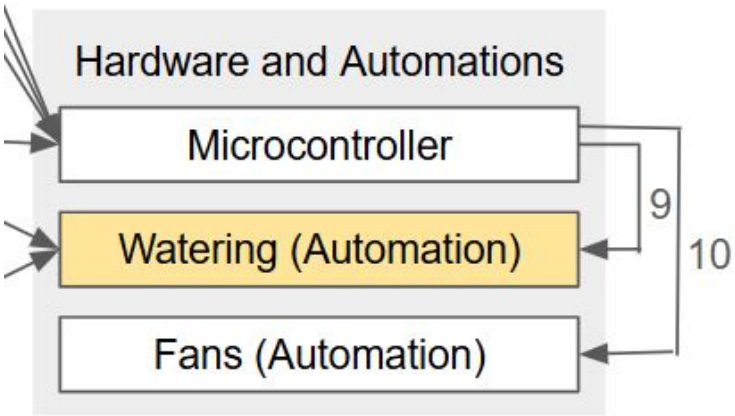

Figure 13: Watering subsystem diagram

6.2.1 Watering (Automation): Hardware

- Water pump
- Piping
- Spots spitters to direct water to the roots

6.2.2 Watering (Automation): Operating System

The system is driven by microcontroller firmware logic and dependent on the Electron JS desktop app for scheduling.

6.2.3 Watering (Automation): Software Dependencies

- Scheduling algorithm that controls watering at certain times of the day and the frequency of it
- Moisture level algorithm
- Manual control algorithm

6.2.4 Watering (Automation): Programming Languages

- C/C++ (Arduino dialect)
- Typescript

6.2.5 Watering (Automation): Data Processing

- useDeviceSchedule() : takes in time, frequency, on and off URLs. Used to set scheduled watering cycles throughout the day.

6.3 Subsystem: Fans (Automation)

This subsystem is responsible for regulating air quality and humidity inside the EGG. When humidity exceeds a set threshold, the fans are triggered to activate for ventilation and dehumidification.

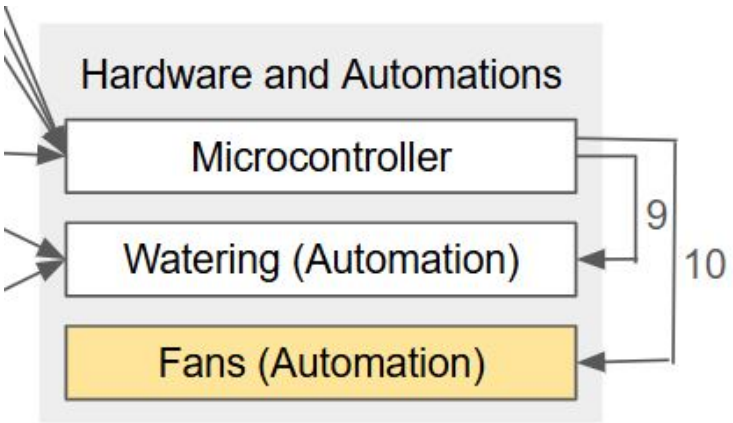

Figure 14: Fans subsystem diagram

6.3.1 Fans (Automation): Hardware

- 5V/12V DC Cooling Fans
- Controlled with a digital pin to a MOSFET, and protected with a Flyback Diode
- Power Supply

6.3.2 Fans (Automation): Operating System

The system is driven by microcontroller firmware logic and dependent on the Electron JS desktop app for scheduling.

6.3.3 Fans (Automation): Software Dependencies

- Moisture level algorithm
- Manual control algorithm

6.3.4 Fans (Automation): Programming Languages

- C/C++ (Arduino dialect)
- Typescript

6.3.5 Fans (Automation): Data Processing

- useDeviceSchedule() : takes in time, frequency, on and off URLs. Used to set scheduled fan cycles throughout the day.

7 Appendix A

Numerous Solidworks and consequentially STL files were used in the assembly and creation of the Encapsulated Gardening Grounds itself. These files enabled the team to create portions of the EGG's assembly out of 3D printed components. Due to the large size of the 3D printed parts of the EGG's assembly, many of the STLs needed to be split into multiple, printable parts. In total, there are around 25 different files used to print the needed components. For brevity sake, a picture of the assembled components can be found below:

- EGGInnerTop1-10.stl: Used to create the inner portion of the moving lid on the EGG.
- EGGOuterTop1-10.stl: Used to create the outer portion of the moving lid on the EGG.
- Coil1-2.stl: Used to create the rotating pin for the moving lid on the EGG.
- EGGTopLip1-2.stl: Used to aesthetically and structurally support the moving lid on the EGG.
- EGGBottomLip1-2.stl: Used to aesthetically and structurally support the top assembly on the EGG.

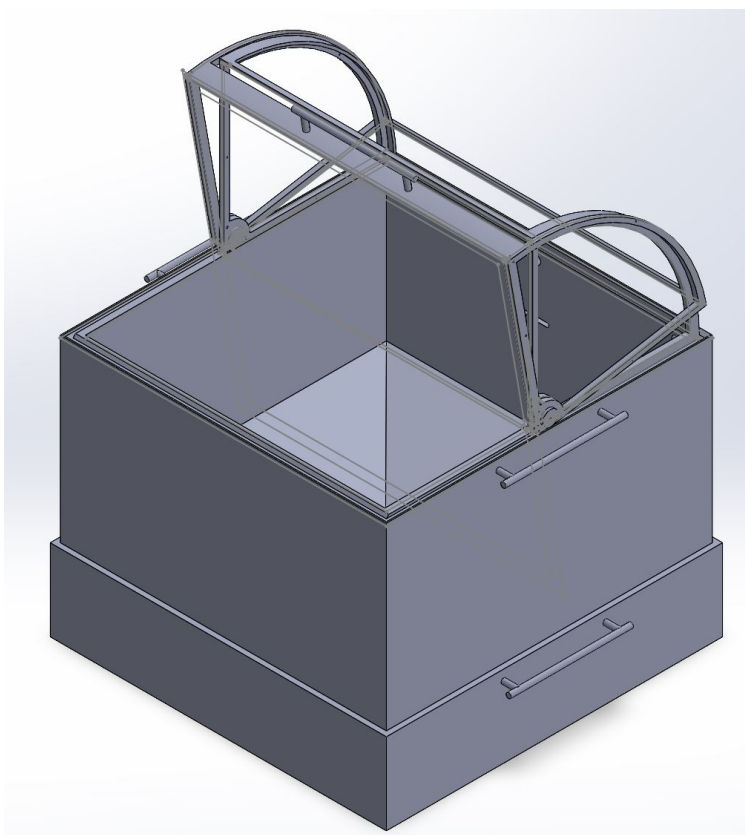

Figure 15: Assembly Diagram